

Android App — Camera, Microphone & WebRTC

1. Permission

Our Android application requests:

- CAMERA → camera access
- RECORD_AUDIO → microphone access

Android grants these permissions to **our application**, not specifically to WebRTC.

2. Calling Flow

```
Android App
  ↓
Request CAMERA + RECORD_AUDIO
  ↓
User grants permission
  ↓
Android Media APIs
  ↓
Camera + Microphone
  ↓
Our WebRTC
  ↓
Internet
  ↓
Other User
```

3. Do We Need Permission Again for WebRTC?

No.

If the Android app already has CAMERA and RECORD_AUDIO permission, the same permission is sufficient when our WebRTC starts.

We only need to **check that the permission is still granted** before starting the call.

There is no separate **"WebRTC permission."**

```
Permission granted
  ↓
Check permission
  ↓
Start WebRTC
```

If the user later revokes the permission, we must request it again.

4. Why Use WebRTC?

Android SDK can provide camera and microphone access, but it does not by itself provide a complete real-time internet calling solution.

Android SDK:

- Permission
- Camera capture
- Microphone capture
- Audio/video playback

WebRTC:

- Real-time audio/video communication
- Peer connection
- Media transmission
- ICE
- STUN/TURN support
- Encryption
- Network handling

5. Third-Party RTC SDK

If we use LiveKit, Agora, Daily, etc.:

```
Our Android App
  ↓
RTC SDK
  ↓
WebRTC / RTC Engine
  ↓
Android Media APIs
  ↓
Camera + Microphone
```

The third-party SDK uses the permissions already granted to **our application**. It does not require a separate WebRTC permission.

Key Takeaway

Android permission belongs to our app. Android Media APIs provide access to camera/microphone. Our WebRTC uses that access to provide real-time communication.